Feature Enhancement: Personalized Insight Pages for ElectricityLoadCalculator.com

Project Context

The website currently includes:

- Appliance-based load calculation
- Total Load (kW/W)
- Daily Consumption
- Monthly Consumption
- Estimated Electricity Bill
- Smart Insights Section
- MCB Recommendation
- Cable Recommendation
- Inverter Recommendation
- Solar Recommendation
- Electrical Health Score
- Consumption Breakdown Analysis
- Energy Saving Recommendations
- PDF Export

The website uses:

- Dark theme
- Amber/yellow accent color
- Mobile-first design
- Card-based UI
- Modern professional appearance

Do NOT redesign the existing visual identity.

Build upon the current Smart Insights cards.

Objective

Transform every Smart Insight card into a fully personalized advisory page.

Currently the site displays:

Recommended MCB: 32A

Recommended Cable: 6 sq mm

Recommended Inverter: 6 kVA

Recommended Solar: 6.5 kW

The new experience should explain:

- Why this recommendation was generated
- How the calculation was performed
- Why alternative options were not selected
- What the recommendation means in practical terms

The goal is to increase:

- User trust
- Time on site
- SEO value
- Internal linking
- Perceived expertise

Global Page Architecture

Every insight page must follow the same structure.

Page Flow:

Personalized Recommendation ↓ Why This Recommendation? ↓ Calculation Breakdown ↓ What This Means ↓ Educational Content ↓ Frequently Asked Questions ↓ Related Recommendations

This structure must remain consistent across all insight pages.

Routes

Create dedicated pages:

/insights/mcb

/insights/cable

/insights/inverter

/insights/solar

Each page must receive calculation results from the report page.

Pass data using:

- Route state
- Query parameters
- Shared state management

Avoid recalculating data unnecessarily.

Reuse existing calculation results whenever possible.

Section 1: Personalized Recommendation Card

At the top of every page display:

Large Recommendation Value

Example:

Recommended MCB

32A C-Curve

Based on your 5.2 kW load

or

Recommended Solar System

6.5 kW

Based on your monthly usage

This section should immediately answer:

"What should I use?"

Section 2: Why This Recommendation?

This is the most important section.

Users must understand how the recommendation was generated.

Display:

Input values

Intermediate calculations

Final recommendation

Use clean cards and step-by-step breakdowns.

MCB Recommendation Page

Top Recommendation

Display:

⚡ Recommended MCB

32A C-Curve

Based on your load

Calculation Breakdown

Show:

Load

5.2 kW

Voltage

230V

Current

5200 ÷ 230

= 22.6A

Safety Margin

22.6 × 1.25

= 28.3A

Nearest Standard Rating

32A

Why Not Smaller?

Explain:

25A MCB

Too close to operating current.

Risk of nuisance tripping.

Why Not Larger?

Explain:

40A MCB

Oversized for current load.

Reduced protection effectiveness.

What This Means

Display:

✓ Suitable for your load

✓ Residential safe sizing

✓ Standard industry recommendation

Educational Content

Include:

- What is an MCB?
- Types of MCBs
- B Curve vs C Curve
- Residential MCB selection

FAQ

Examples:

What happens if MCB size is too small?

What happens if MCB size is too large?

Which MCB curve is best for homes?

Cable Recommendation Page

Top Recommendation

🔌 Recommended Cable

6 sq mm Copper

Calculation Breakdown

Display:

Recommended MCB

32A

Required Current

28.3A

Selected Cable

6 sq mm Copper

Cable Selection Table

Example:

1.5 sq mm → 10A

2.5 sq mm → 16A

4 sq mm → 25A

6 sq mm → 32A ✓

10 sq mm → 40A

Highlight selected size.

Why This Cable?

Explain:

- Current carrying capacity
- Safety margin
- Reduced overheating risk
- Suitable residential performance

Educational Content

Include:

- Copper vs Aluminium
- Common residential cable sizes
- Voltage drop basics
- Cable safety guidelines

FAQ

Examples:

Can I use a smaller cable?

Why is copper preferred?

What causes cable overheating?

Inverter Recommendation Page

Top Recommendation

🔋 Recommended Inverter

6 kVA

Calculation Breakdown

Display:

Load

5.2 kW

Sizing Factor

1.2

Calculation

5.2 × 1.2

= 6.24

Selected Capacity

6 kVA

Backup Estimation

Display examples:

2 Batteries

≈ 3 Hours Backup

4 Batteries

≈ 6 Hours Backup

What This Means

Explain:

✓ Suitable for your connected load

✓ Handles startup demand

✓ Allows future expansion

Educational Content

Include:

- Difference between kW and kVA
- Inverter sizing basics
- Battery selection
- Backup planning

FAQ

Examples:

What inverter size do I need?

Can a smaller inverter work?

How is backup time calculated?

Solar Recommendation Page

Top Recommendation

☀️ Recommended Solar System

6.5 kW

Calculation Breakdown

Display:

Monthly Units

753

Solar Production Factor

120 units per kW per month

Calculation

753 ÷ 120

= 6.27

Recommended System

6.5 kW

Estimated Benefits

Display:

Monthly Savings

₹4,500

Annual Savings

₹54,000

Estimated Payback

4–6 Years

What This Means

Explain:

✓ Can offset most electricity consumption

✓ Reduce monthly electricity bill

✓ Long-term cost savings

Educational Content

Include:

- Net metering
- Solar sizing basics
- Rooftop requirements
- Solar ROI

FAQ

Examples:

How much roof area is needed?

What is net metering?

How long does solar last?

Related Recommendations Section

At the bottom of every page display:

Related Insights

Example:

Recommended Cable

Recommended Inverter

Recommended Solar

Recommended MCB

Allow easy navigation between all recommendation pages.

SEO Requirements

Each page must contain:

- User-specific recommendation
- Dynamic calculation explanation
- Educational content
- FAQ section
- Internal links

The page should be useful both:

1. For calculator users.
2. For users landing directly from Google.

UX Requirements

Maintain existing:

- Dark theme
- Yellow accent styling
- Card-based layout
- Mobile-first experience

The recommendation and calculation sections must appear above educational content.

Users should immediately see:

1. The recommendation.
2. Why it was recommended.
3. How it was calculated.

before scrolling into educational content.

Success Criteria

Every insight page should feel like a professional engineering recommendation report rather than a generic blog article.

Users should leave the page understanding:

✓ What was recommended

✓ Why it was recommended

✓ How it was calculated

✓ What action they should take next

The final experience should increase trust, engagement, session duration, internal page navigation, and SEO authority across the entire website.